Finding molecules in cells — fast

2D template matching for in situ structural biology

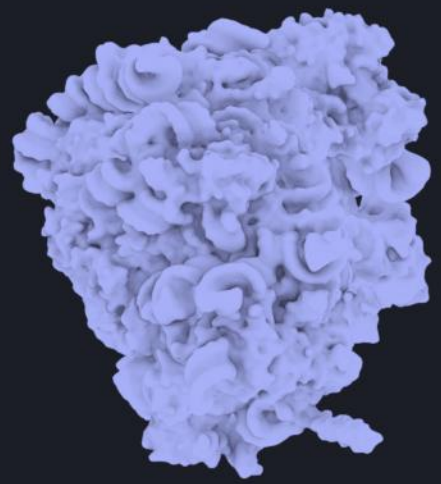
Detect · Decode · Discover

Kexin Zhang · Computational Imaging and Discovery Lab

kexinzhang-lab.github.io · SHM CE11

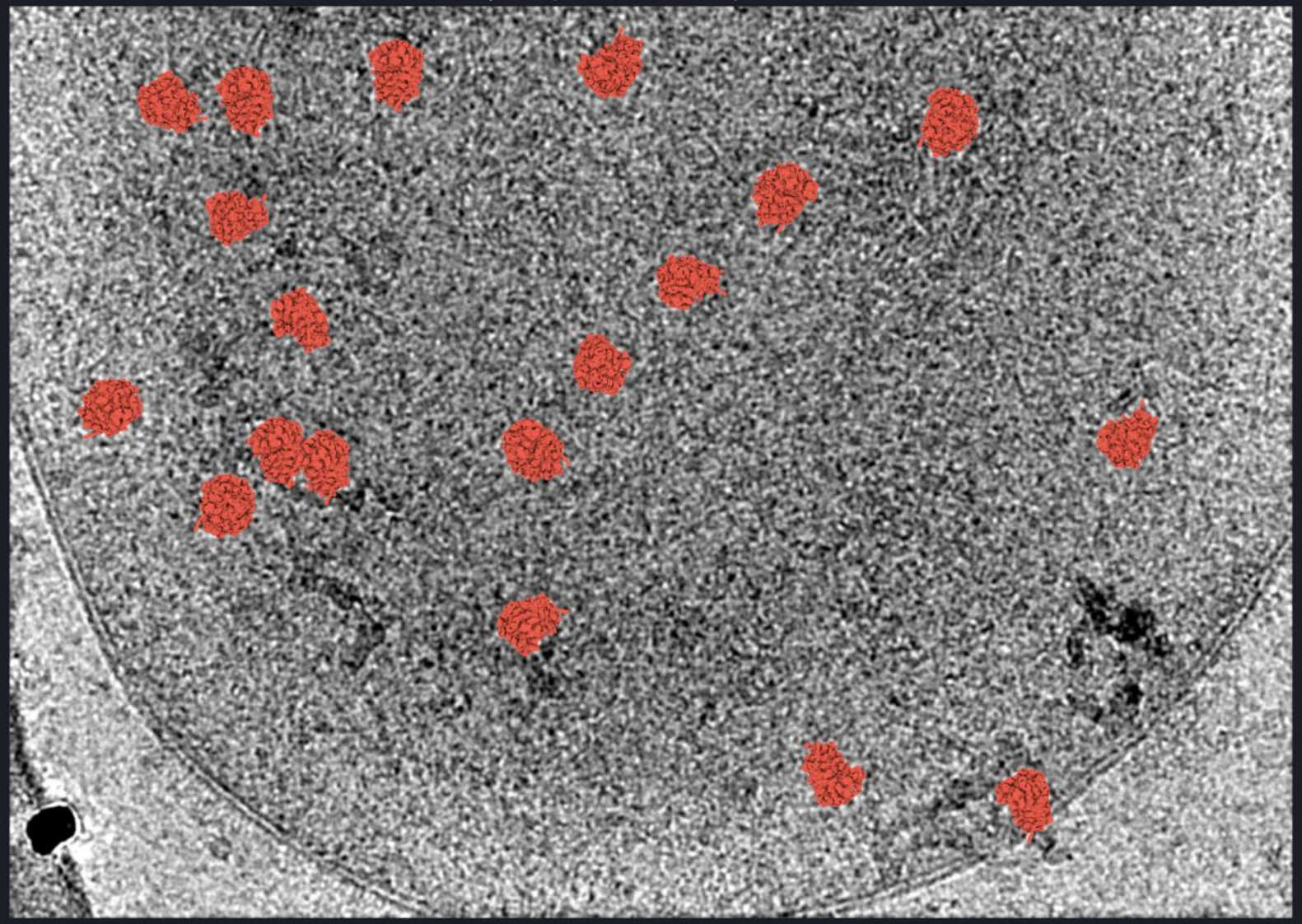
Template (3D map)

60S ribosomal subunit (PDB 5LZV)



Micrograph

BSC-1 cell lysate (control, with FBS) · EMPIAR-13577



Data: Seraj Z, et al. eLife 2026

our fast 2DTM, Accuracy mode: 16 true + 3 false positives

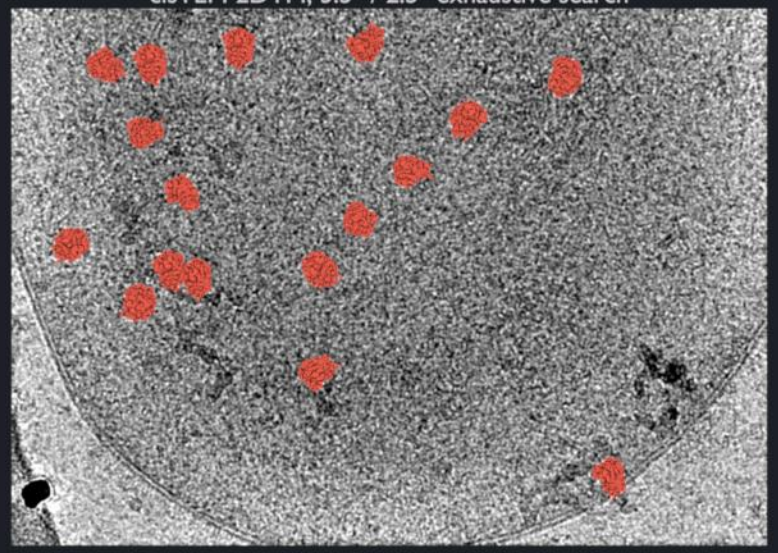
detections: 19

Near real-time 2D template matching

BSC-1 cell lysate (control, with FBS) · EMPIAR-13577 · template: 60S ribosomal subunit (PDB 5LZV)

Reference

cisTEM 2DTM, 3.5° / 2.5° exhaustive search



17 reference particles

140 s on RTX 5080 (our fast cisTEM build)

525 s on RTX 5080 (stable cisTEM build)

true positive false positive missed reference

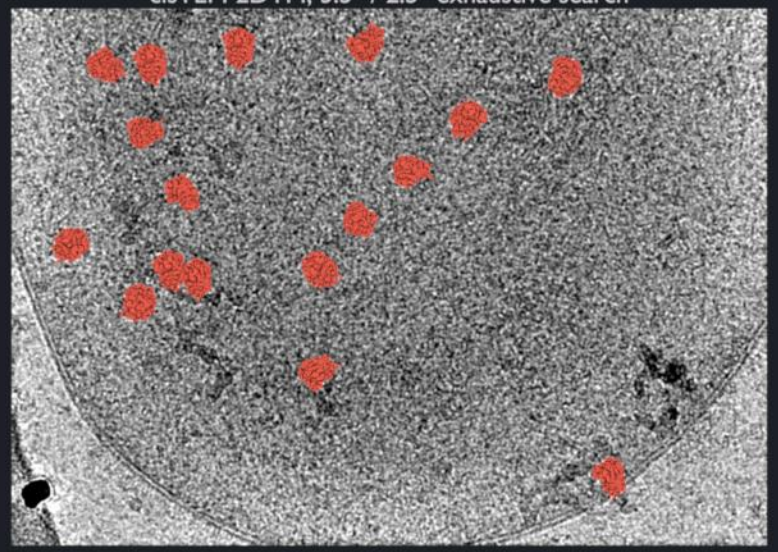
Data: Seraj Z, et al. eLife 2026

Near real-time 2D template matching

BSC-1 cell lysate (control, with FBS) · EMPIAR-13577 · template: 60S ribosomal subunit (PDB 5LZV)

Reference

cisTEM 2DTM, 3.5° / 2.5° exhaustive search



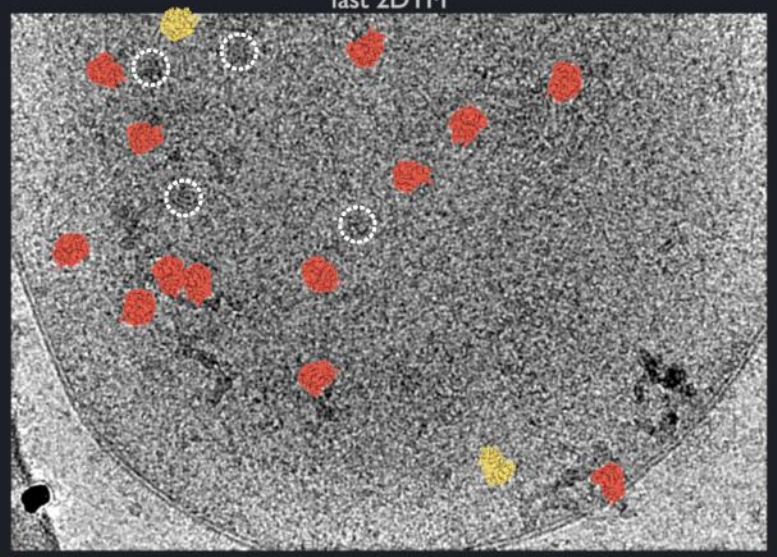
17 reference particles

140 s on RTX 5080 (our fast cisTEM build)

525 s on RTX 5080 (stable cisTEM build)

Speed mode

fast 2DTM



TP 13 FP 2 missed 4

recall 76% precision 87%

10.4 s on RTX 5080

true positive · false positive · missed reference

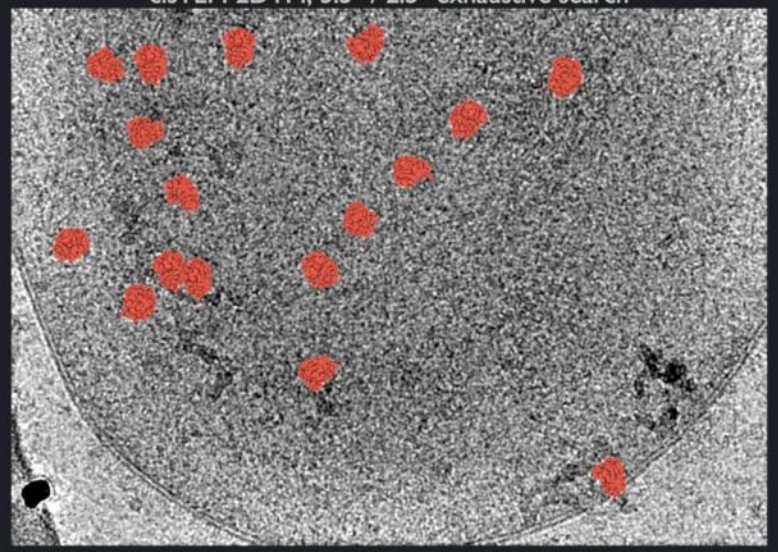
Data: Seraj Z, et al. eLife 2026

Near real-time 2D template matching

BSC-1 cell lysate (control, with FBS) · EMPIAR-13577 · template: 60S ribosomal subunit (PDB 5LZV)

Reference

cisTEM 2DTM, 3.5° / 2.5° exhaustive search



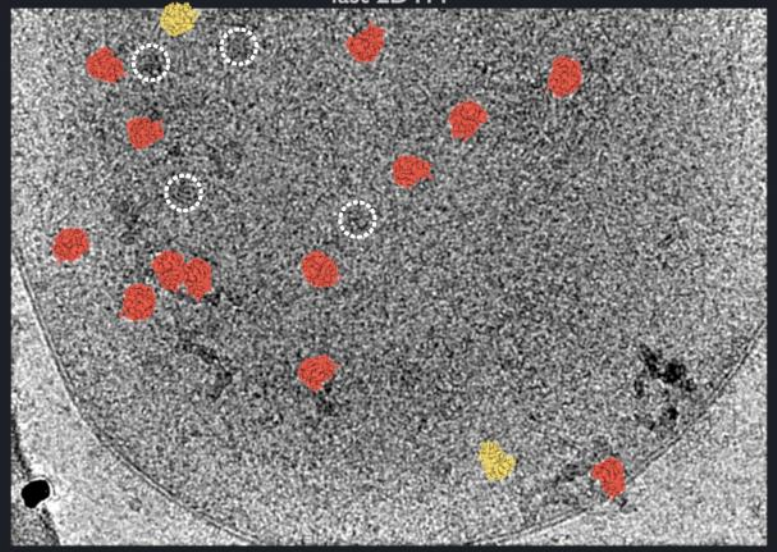
17 reference particles

140 s on RTX 5080 (our fast cisTEM build)

525 s on RTX 5080 (stable cisTEM build)

Speed mode

fast 2DTM



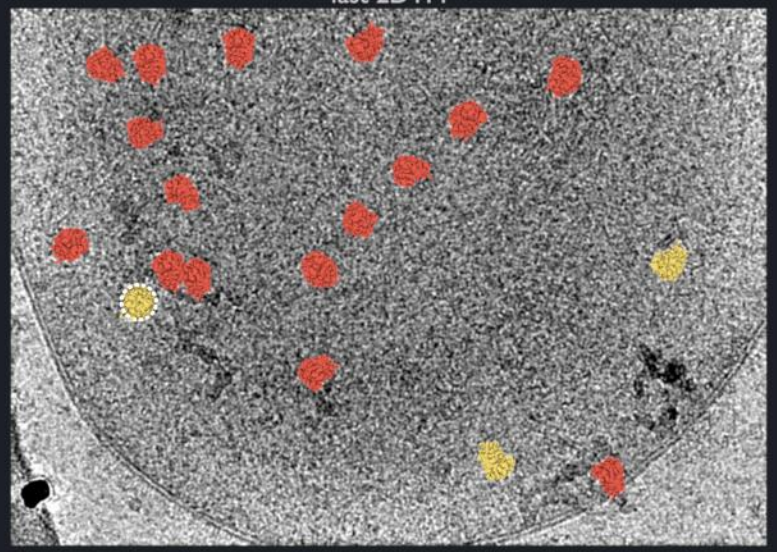
TP 13 FP 2 missed 4

recall 76% precision 87%

10.4 s on RTX 5080

Accuracy mode

fast 2DTM



TP 16 FP 3 missed 1

recall 94% precision 84%

30.2 s on RTX 5080

true positive · false positive · missed reference

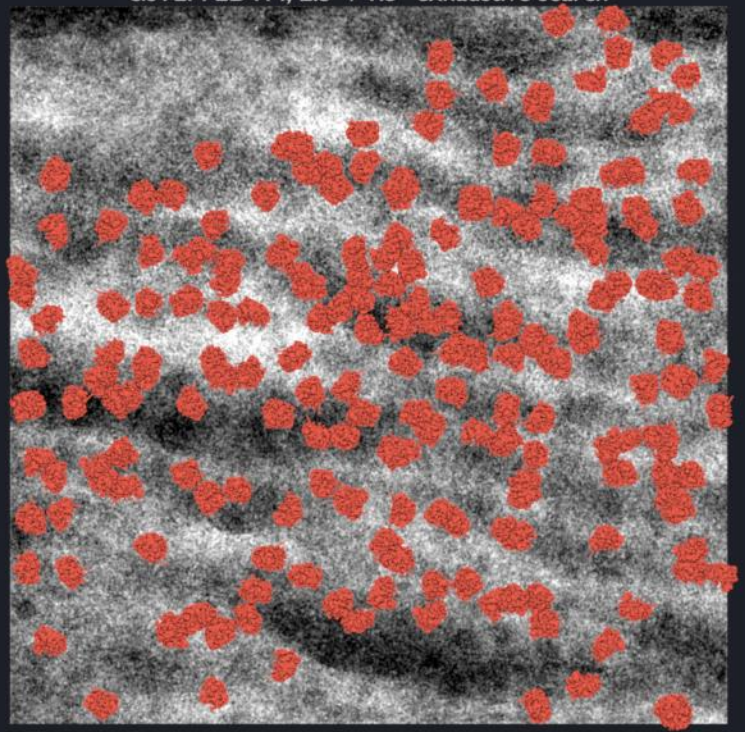
Data: Seraj Z, et al. eLife 2026

Near real-time 2D template matching in situ

CEMOVIS section of S. cerevisiae cells · EMPIAR-12835 · template: 60S ribosomal subunit (PDB 6Q8Y)

Reference

cisTEM 2DTM, 2.5° / 1.5° exhaustive search



206 reference particles

36 min on RTX 5080 (our fast cisTEM build)

stable cisTEM build: TBD

true positive false positive missed reference

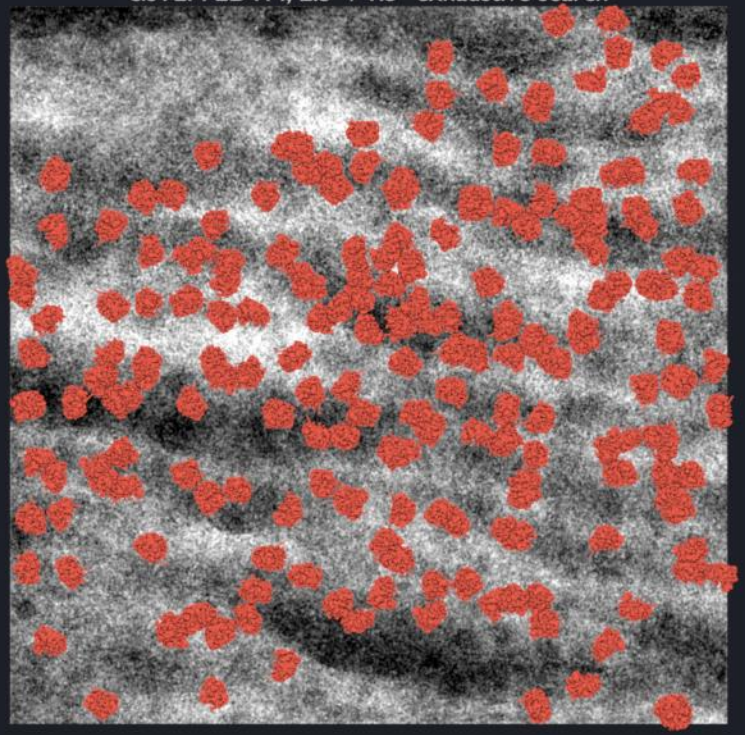
Data: Elferich J, et al. IUCrJ 2025

Near real-time 2D template matching in situ

CEMOVIS section of S. cerevisiae cells · EMPIAR-12835 · template: 60S ribosomal subunit (PDB 6Q8Y)

Reference

cisTEM 2DTM, 2.5° / 1.5° exhaustive search



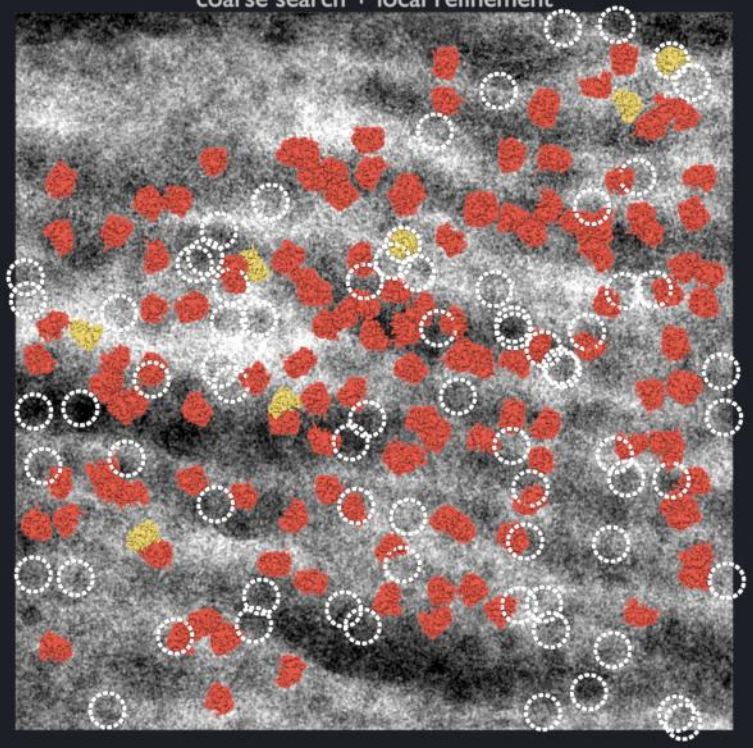
206 reference particles

36 min on RTX 5080 (our fast cisTEM build)

stable cisTEM build: TBD

Our fast 2DTM

coarse search + local refinement



TP 134 FP 7 missed 72

recall 65% precision 95%

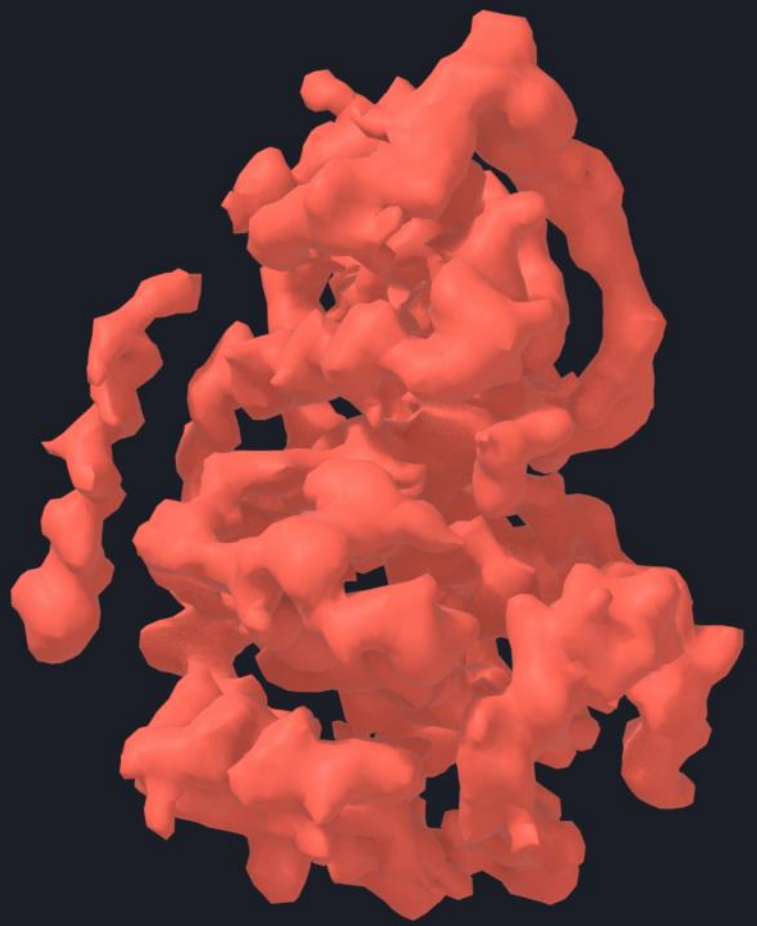
26.6 s on RTX 5080

true positive · false positive · missed reference

Data: Elferich J, et al. IUCrJ 2025

Composite omit map

each voxel from the reconstruction whose template lacked it



PKA (PDB 1ATP) · EMPIAR-10252
Data: Herzik MA Jr, et al. Nat Commun 2019 · Method: Zhang K, et al. eLife 2026

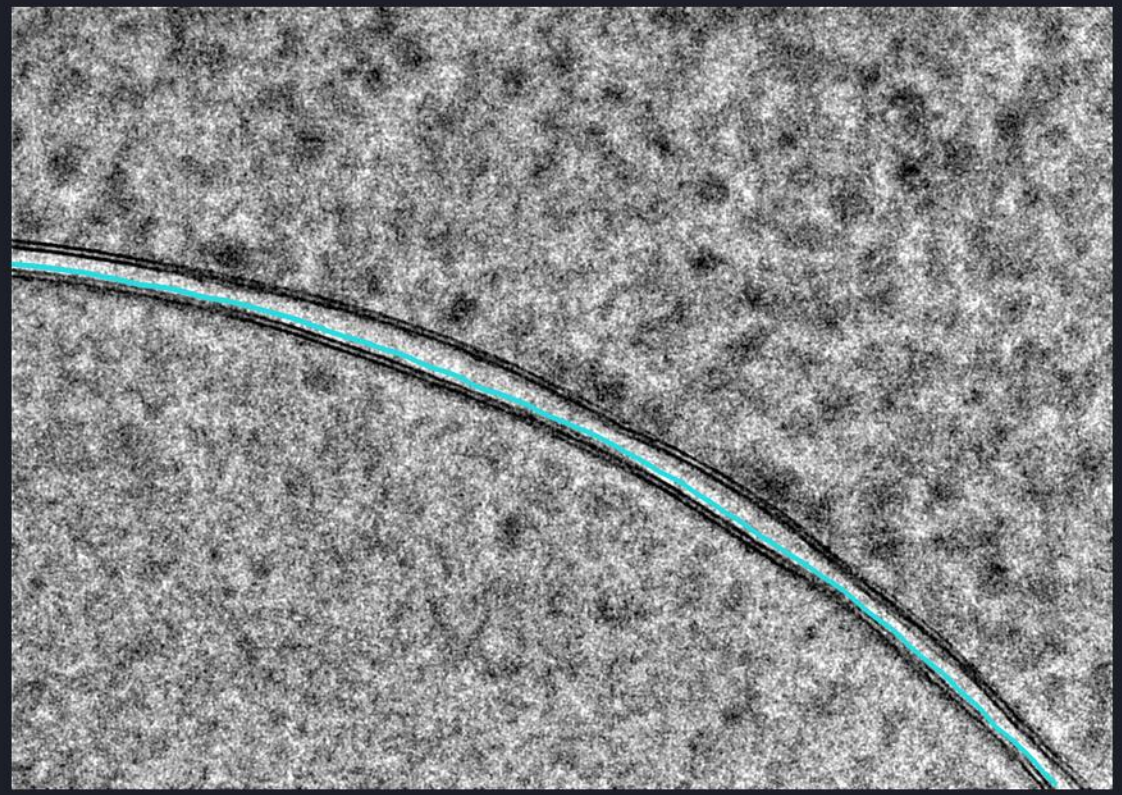
omit templates: 36 / 36

Toward context-aware molecular detection

learn cellular morphology (membranes, flagella) to know where molecules sit

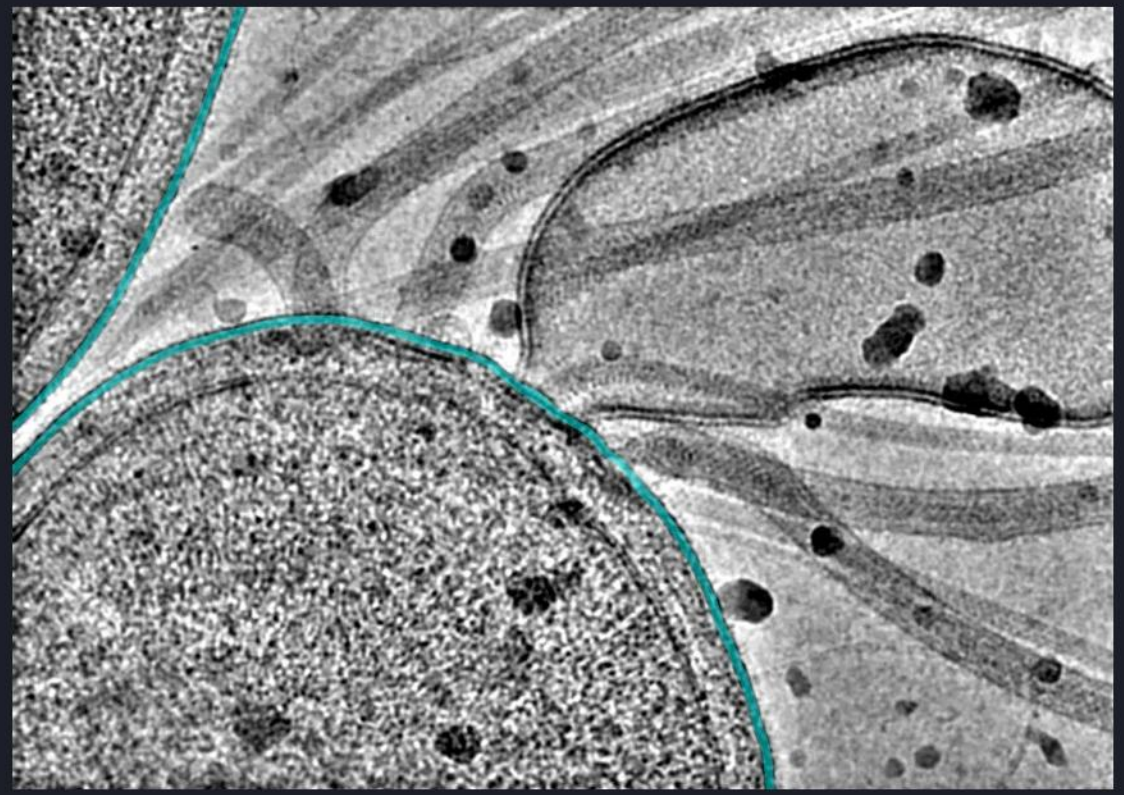
Yeast lamella · numerical method

membrane tracing without training



Bacterial cell · CNN (U-Net)

trained on annotated cryo-EM images





Hang Li
CNN extension to bacteria



Wangbiao Guo
Jun Liu lab, Yale · collaboration

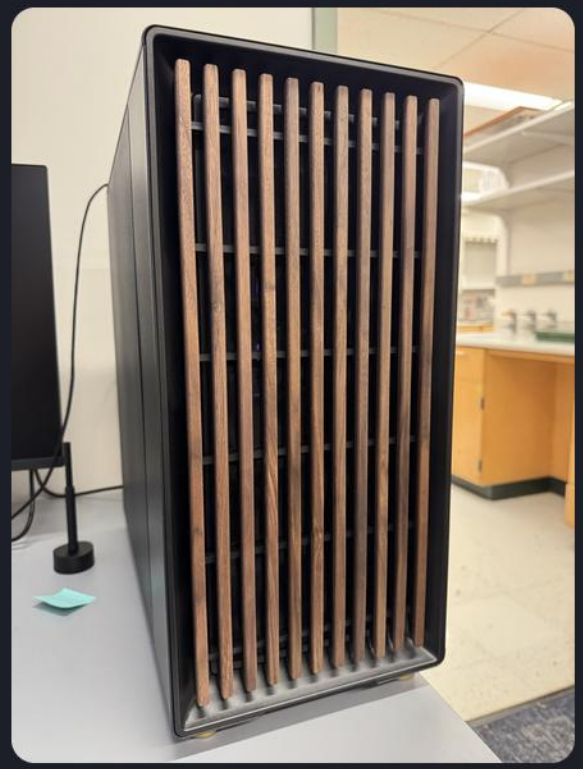
numerical + CNN segmentation → 2DTM detections in their cellular context

Yeast data: EMPIAR-10998, Lucas BA, et al. eLife 2022

Rotations open — come chat with us

about science, GPUs, and coffee







Our GPU workstation, built in-house by Hang Li

the RTX 5080 behind today's timings



The coffee nook

drop by SHM CE11 any time

2D template matching · in situ structure · context-aware molecular detection

Computational Imaging and Discovery Lab · Kexin Zhang · SHM CE11 · kexinzhang-lab.github.io